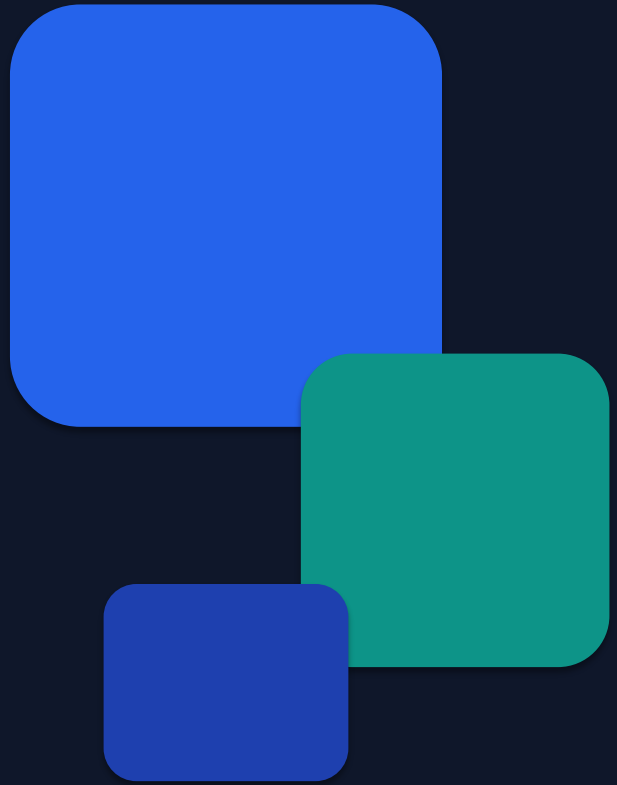
DBMS • SEMESTER 5 MINI PROJECT

INSURANCE CLAIMS PROCESSING SYSTEM

A relational database and web-based application



PROJECT TEAM

Shreya Shenoy | PES2UG24CS487
Shreya Patil | PES2UG24CS484

TEAM 1 - H section

Problem & Proposed Solution

THE PROBLEM

Insurance processing involves connected records across customers, policies, incidents, claims, assessments, settlements and payments. Fragmented records make tracking and retrieval difficult.

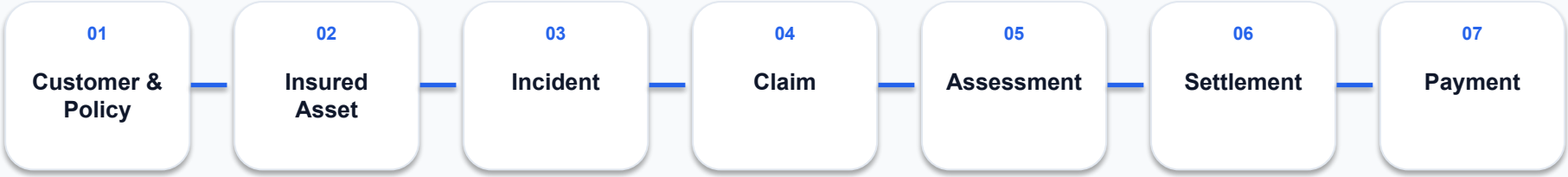
OUR APPROACH

A structured MySQL database links the major entities through keys and constraints. A web interface provides access to relevant records through application APIs.

PROJECT OBJECTIVES

- Design a relational model for the insurance claim lifecycle.
- Maintain integrity using primary keys, foreign keys and constraints.
- Demonstrate SQL queries and advanced database features.
- Integrate the database with a web-based application.

System Scope & Claim Workflow



PROJECT SCOPE

- Customer and policy records
- Asset and incident information
- Claims, assessments and settlements
- Payment records and SQL analysis

APPLICATION LAYER

- Next.js / React interface
- API routes for database operations
- MySQL connectivity using mysql2
- Dashboard and record modules

Relational Database Design

10 ENTITIES ORGANISE THE INSURANCE DOMAIN

CUSTOMER
Customer details

POLICY_TYPE
Policy categories

INSURANCE_POLICY
Customer policies

INSURED_ASSET
Covered assets

INCIDENT
Incident details

CLAIM
Submitted claims

ASSESSOR
Assessor details

CLAIM_ASSESSMENT
Claim evaluations

CLAIM_SETTLEMENT
Settlement records

PAYMENT
Payment records

Key Entity Relationships

CUSTOMER	1 : N	INSURANCE_POLICY
POLICY_TYPE	1 : N	INSURANCE_POLICY
INSURANCE_POLICY	1 : 1	INSURED_ASSET
INSURANCE_POLICY	1 : N	CLAIM
CLAIM	1 : 1	CLAIM_ASSESSMENT
ASSESSOR	1 : N	CLAIM_ASSESSMENT
CLAIM	1 : 0..1	CLAIM_SETTLEMENT
CLAIM_SETTLEMENT	1 : 0..1	PAYMENT

* Cardinalities reflect our current schema constraints and optional records.

SQL Implementation

DDL

CREATE DATABASE / TABLE
PRIMARY KEY & FOREIGN KEY
ALTER TABLE • VIEW • INDEX

DML

INSERT • SELECT
UPDATE • DELETE
Sample data across 10 tables

QUERYING

JOIN • GROUP BY • HAVING
Aggregate functions
Nested & correlated subqueries

TRANSACTIONS

START TRANSACTION
COMMIT • ROLLBACK
Staging-table demonstration

Purpose: define the schema, manipulate records, retrieve insights and demonstrate controlled database operations.

Advanced Database Features

STORED PROCEDURE

GetClaimDetails
Retrieves consolidated information for a specified claim.

STORED FUNCTION

CalculateSettlementPercentage
Returns settlement as a percentage of the claimed amount.

TRIGGER

after_claim_settlement
Updates claim status after a settlement is inserted.

VIEW

vw_claim_summary
Combines claim, customer, policy and policy type information.

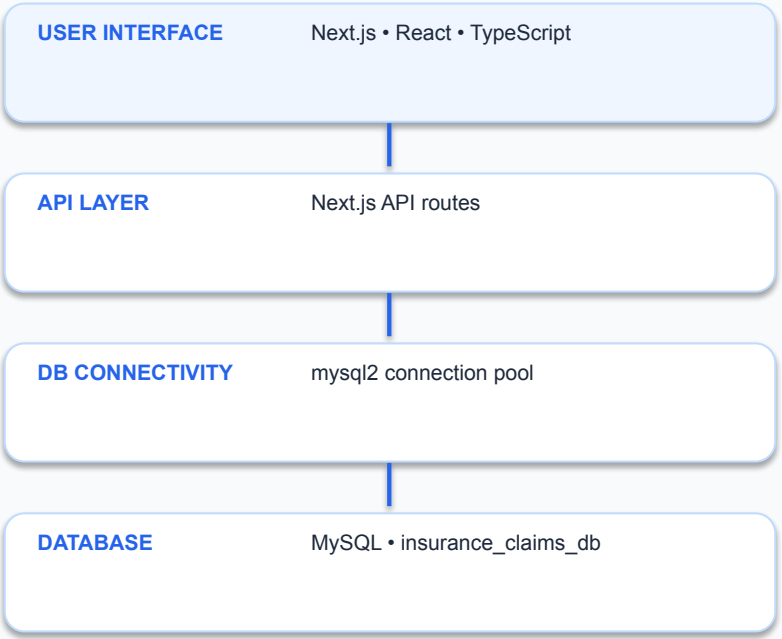
COMPOSITE INDEX

idx_claim_status_date
Index on claim_status and claim_date.

TRANSACTIONS

COMMIT and ROLLBACK
Demonstrates saving and reversing uncommitted changes.

Web Application & Architecture



APPLICATION MODULES

- Admin Dashboard – summary counts
- Customers – records and search
- Policies – policy information
- Claims – claim information

DEVELOPMENT TOOLS

Tailwind CSS for styling • Environment variables for configuration
GitHub Copilot and OpenAI Codex assisted frontend development.

Validation & Live Demonstration

DATABASE CHECKS

Verify database and tables
Retrieve sample customer, policy and claim records
Run selected SQL queries and routines

APPLICATION CHECKS

Open dashboard and completed modules
Demonstrate live record retrieval
Show search and API responses

SAMPLE DATA COUNTS

Customers	Policies	Claims	Settlements
6	6	6	3

Counts reflect the sample-data verification reported during development; confirm live values during the demo.

Conclusion & Future Enhancements

WHAT WE ACHIEVED

Designed a ten-table relational model
Implemented SQL and advanced database features
Connected MySQL to a web application
Demonstrated retrieval and data relationships

POSSIBLE ENHANCEMENTS

Extend role-based access
Expand claim workflow automation
Add richer reporting and analytics
Complete or extend NoSQL integration as applicable

Thank you

References

- MySQL Documentation — https://dev.mysql.com/doc/
- Next.js Documentation — https://nextjs.org/docs
- React Documentation — https://react.dev/
- Tailwind CSS Documentation — https://tailwindcss.com/docs
- GitHub Copilot Documentation — https://docs.github.com/en/copilot
- OpenAI Codex Documentation — https://developers.openai.com/codex/
- PES University — DBMS Semester 5 Mini Project Guidelines and Lab Materials

Project repository: github.com/shenoyshre22/Insurance-Claims-database-